Health & Community Impacts of Data Centers

Six ways a data center affects the people who live near one — the health risks plus higher bills and the environment. Every claim sourced on the pages that follow, with the permit condition that addresses it.

Air pollution

Fossil power plants and diesel backup generators serving data centers emit pollution linked to ~1,300 premature deaths a year by 2030.

Noise pollution

Cooling systems and generators produce a constant low-frequency hum linked to sleep disturbance, hypertension, and cardiovascular disease.

Light pollution

24/7 campus lighting disrupts sleep and circadian rhythms; the AMA has warned about nighttime light exposure since 2012.

Higher bills

Large loads strain the grid and shift transmission and capacity costs onto residential ratepayers.

Water consumption

Data centers consume water twice — on site for cooling and at the power plants that supply them — often in water-stressed regions.

Climate & reliability

Fossil-heavy demand growth raises emissions, strains grid reliability, and adds local waste heat.

AIR POLLUTION

- UC Riverside & Caltech researchers project up to ~1,300 premature deaths a year by 2030 from U.S. data-center air pollution — that is a midpoint, with a published range of 940 to 1,590 — alongside roughly 600,000 asthma-symptom cases, and a total public-health burden exceeding $20B in 2028. Quote the range rather than the midpoint: it is the defensible version, and a point estimate invites an argument about precision you do not need to have.

  Source: Han, Wu, Li, Wierman & Ren — The Unpaid Toll: Quantifying the Public Health Impact of AI (arXiv:2412.06288, UC Riverside/Caltech, 2024)
  https://arxiv.org/abs/2412.06288

- The pollution comes from two places: the fossil-fueled power plants supplying the electricity and on-site diesel backup generators emitting NOx and fine particulates (PM2.5) — both linked to respiratory disease, cardiovascular conditions, and cancer risk.

  Source: Han, Wu, Li, Wierman & Ren — The Unpaid Toll: Quantifying the Public Health Impact of AI (arXiv:2412.06288, UC Riverside/Caltech, 2024)
  https://arxiv.org/abs/2412.06288

- The burden lands unevenly: much of the health impact falls on communities near the power plants that serve a data center, which can sit hundreds of miles from the facility itself.

  Source: Han, Wu, Li, Wierman & Ren — The Unpaid Toll: Quantifying the Public Health Impact of AI (arXiv:2412.06288, UC Riverside/Caltech, 2024)
  https://arxiv.org/abs/2412.06288

WHAT TO DEMAND

Demand generator run-hour limits, Tier 4 (or battery) backup instead of legacy diesel, and an air-permit review with public comment before any approval.

NOISE POLLUTION

- Data-center cooling systems and backup generators produce a continuous low-frequency hum that travels farther than ordinary sound and penetrates building walls.

  Source: Environmental Health Project — The Health Risks of Data Centers (infographic, 2026)
  https://www.environmentalhealthproject.org/_files/ugd/a9ce25_3c3574ea12324c65909d308c3a716e56.pdf

- The WHO links chronic environmental noise to sleep disturbance, cardiovascular and metabolic disease, and cognitive impairment — in Europe, environmental noise contributes to an estimated 48,000 new heart-disease cases and 12,000 premature deaths every year.

  Source: European Environment Agency — Health risks caused by environmental noise in Europe (2020)
  https://www.eea.europa.eu/en/analysis/publications/health-risks-caused-by-environmental-noise-in-europe

- WHO night-noise recommendations sit around 40-45 dB — the reason our model CBA clause sets 45 dBA at the nearest residential property line.

  Source: WHO Europe — Environmental Noise Guidelines: noise & cardiovascular/metabolic mechanisms (2018)
  https://www.who.int/europe/publications/i/item/WHO-EURO-2018-3009-42767-59666

WHAT TO DEMAND

A 45 dBA limit at the residential property line as a permit condition — measured after commissioning, not just modeled — with quarterly public reporting.

LIGHT POLLUTION

- The American Medical Association adopted policy in 2012 recognizing nighttime light exposure as a health hazard: it suppresses melatonin and disrupts circadian rhythms.

  Source: American Medical Association — Council on Science & Public Health report on light pollution & nighttime lighting (2012 policy)
  https://www.ama-assn.org/sites/ama-assn.org/files/corp/media-browser/public/about-ama/councils/Council%20Reports/council-on-science-public-health/a12-csaph4-lightpollution-summary.pdf

- Chronic light at night is associated with reduced sleep, impaired daytime functioning, obesity, and mood disorders, with research linking elevated light pollution to higher breast and prostate cancer rates.

  Source: American Medical Association — Council on Science & Public Health report on light pollution & nighttime lighting (2012 policy)
  https://www.ama-assn.org/sites/ama-assn.org/files/corp/media-browser/public/about-ama/councils/Council%20Reports/council-on-science-public-health/a12-csaph4-lightpollution-summary.pdf

- Data center campuses typically run high-intensity exterior security lighting around the clock.

  Source: Environmental Health Project — The Health Risks of Data Centers (infographic, 2026)
  https://www.environmentalhealthproject.org/_files/ugd/a9ce25_3c3574ea12324c65909d308c3a716e56.pdf

WHAT TO DEMAND

Full-cutoff shielded fixtures, color temperature at or below 3000K (the AMA community guidance), and dark-sky-compliant site lighting as approval conditions.

HIGHER BILLS

- PJM's 2025 capacity auction cleared at a record $16.4B, of which the market monitor attributes $6.3B — 38% — to data centers. Across PJM's last four base auctions it is $29.4B of $63.6B, or 46%. These costs flow through to ratepayers across 13 states.

  Source: Monitoring Analytics (PJM market monitor) — data centers drove $6.3B of the record $16.4B 2025 capacity auction (38%), and $29.4B of $63.6B across the last four auctions (46%); see the 2025/2026 RPM Base Residual Auction analyses
  https://www.monitoringanalytics.com/reports/Reports/2025.shtml

- Virginia's legislative audit agency (JLARC) found that unconstrained data-center growth will raise costs for other customers absent policy changes — in the state with more data centers than any other.

  Source: Virginia JLARC — Data Center Impact Study (Report 591, Dec 2024)
  https://jlarc.virginia.gov/pdfs/reports/Rpt591.pdf

- Berkeley economists warn that who pays for data-center-driven grid expansion is a policy choice: without large-load tariffs, the default is everyone.

  Source: UC Berkeley Energy Institute — What Will Data Centers Do To Your Electric Bill? (2025)
  https://energyathaas.wordpress.com/2025/09/29/what-will-data-centers-do-to-your-electric-bill/

WHAT TO DEMAND

Cost causation as a condition: the developer pays 100% of interconnection and grid upgrades, under a large-load tariff so costs never reach residential bills. Estimate your own exposure in the Local Impact Calculator.

WATER CONSUMPTION

- About one-fifth of data centers' direct water footprint is drawn from moderately-to-highly water-stressed watersheds, and nearly half of servers are powered by plants in water-stressed regions.

  Source: Siddik, Shehabi & Marston — The environmental footprint of data centers in the United States (Environmental Research Letters, 2021)
  https://iopscience.iop.org/article/10.1088/1748-9326/abfba1

- Water is consumed twice: directly for evaporative cooling on site, and indirectly at the thermoelectric power plants generating the electricity.

  Source: Siddik, Shehabi & Marston — The environmental footprint of data centers in the United States (Environmental Research Letters, 2021)
  https://iopscience.iop.org/article/10.1088/1748-9326/abfba1

- AI-specific demand is growing fast — researchers project AI's water withdrawal could reach billions of cubic meters a year by 2027, and most operators still don't report site-level water use.

  Source: Li et al., Making AI Less Thirsty — water footprint of AI (arXiv:2304.03271)
  https://arxiv.org/abs/2304.03271

WHAT TO DEMAND

An enforceable annual water cap in the permit, recycled or non-potable cooling supply, quarterly public metering, and re-approval before expansion.

CLIMATE & RELIABILITY

- U.S. data-center electricity use is projected to reach 325-580 TWh by 2028 (Berkeley Lab) — 6.7% to 12% of national consumption, up from 176 TWh and 4.4% in 2023 — with much of the new supply coming from natural gas.

  Source: Lawrence Berkeley National Lab, 2024 US Data Center Energy Usage Report — 325–580 TWh (6.7–12% of US electricity) by 2028, up from 176 TWh (4.4%) in 2023
  https://eta.lbl.gov/publications/2024-united-states-data-center-energy

- The IEA projects data centers will drive one of the largest sources of electricity demand growth this decade, with the fuel mix determining the emissions impact.

  Source: IEA, Energy and AI (2025) + Key Questions update (2026)
  https://www.iea.org/reports/energy-and-ai

- Hyperscalers are increasingly turning to on-site gas generation as a 'bridge' — locking in fossil combustion next to host communities.

  Source: Natural Gas Intelligence — On-site natural gas generation gains favor with hyperscalers (2026)
  https://naturalgasintel.com/news/on-site-natural-gas-generation-gains-favor-with-hyperscalers-as-bridge-to-grid/

WHAT TO DEMAND

Binding 24/7 carbon-free energy commitments, no ratepayer-funded gas buildout, and demand-response/curtailment agreements so the facility sheds load in grid emergencies.